



CHIJ ST. THERESA'S CONVENT
PRELIMINARY EXAMINATION 2026
SECONDARY 4 EXPRESS / 5 NORMAL (ACADEMIC)

CANDIDATE
NAME

CLASS

INDEX
NUMBER

ADDITIONAL MATHEMATICS

4049/2

Paper 2

25 August 2026

2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an approved scientific calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 90.

Mathematical Formula

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!} = \frac{n(n-1)\dots(n-r+1)}{r!}$.

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2}ab \sin C$$

1 Show that the line $y = 4x$ intersects the curve $y = x^2 + kx + 3 - k$ for all real values of k . [5]

2 The curve $y = x\sqrt{x+1}$ has a stationary value at the point P . Find the coordinates of P . [6]

- 3 (a) The expression $f(x) = ax^3 + bx^2 - 7x + 6$, where a and b are constants, has a factor of $x + 1$ and it leaves a remainder of $\frac{3}{2}$ when divided by $2x - 1$. Find the values of a and b . [4]

- (b) Solve the equation $4x^3 - 9x^2 + 4 = 0$, expressing the roots in an exact form. [5]

- 4 A quadrilateral is formed by the four points (a, a^3) , $(-b, -b^3)$, $(1, 1)$ and $(-1, -1)$, where the constants a and b are such that $a > b > 1$. Express the area of the quadrilateral as a product of two factors, one of which is $(a + b)$. [4]

- 5 Solve the equation $\lg x^3 = 1 - \log_x(0.0001)$. [6]

6 The equation of a curve is $y = e^{-\frac{1}{2}x} \cos x$.

(i) Find $\frac{dy}{dx}$. [2]

The curve intersects the y -axis at the point A .

(ii) Find the gradient of the normal to the curve at the point. [2]

The smallest positive value of x at which the curve has a stationary point is $x = x_0$.

(iii) Find the value of x_0 . [3]

7 (a) Prove the identity $\tan\left(x + \frac{\pi}{4}\right) = \frac{1 + \sin 2x}{\cos 2x}.$ [4]

- (b) Hence solve the equation $1 + \sin 2x = 3 \cos 2x$ for $-2\pi \leq x \leq 2\pi$. [6]

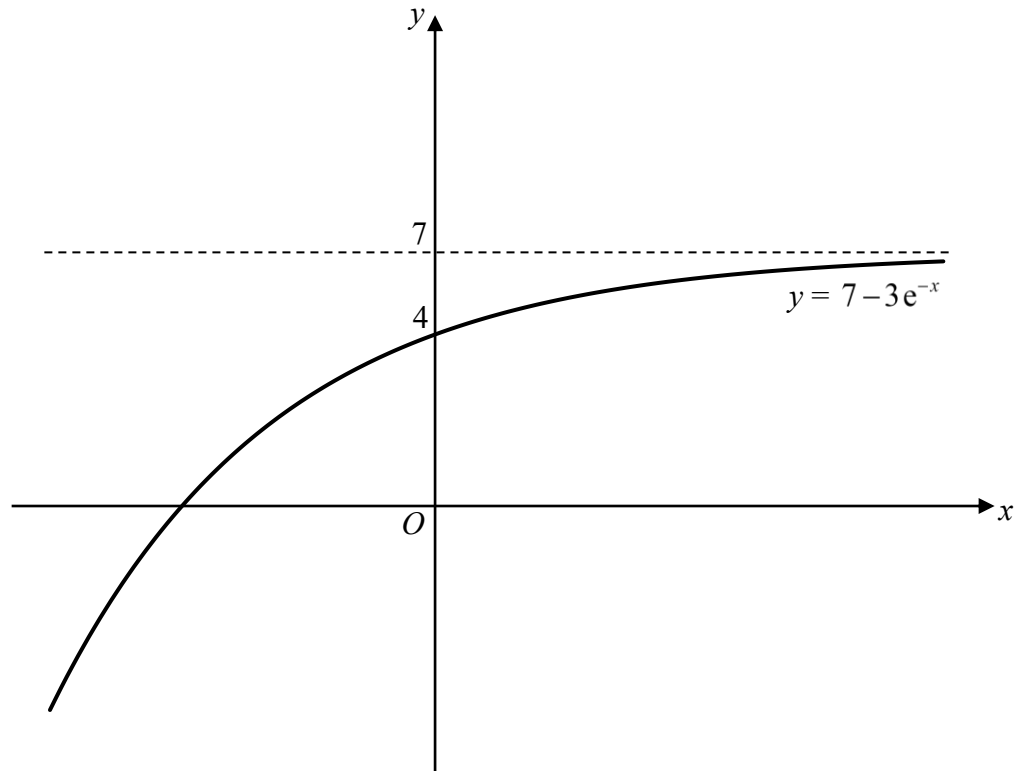
8 The equation of a curve is $y = \sec x - \tan x$.

(i) Show that $\frac{dy}{dx} = -\left(\frac{1}{1 + \sin x}\right)$. [5]

The line L is a tangent to the curve at the point where $x = -\frac{\pi}{6}$.

(ii) Find the y -intercept of L . [3]

- 9 The diagram shows the curve of $y = 7 - 3e^{-x}$.



- (i) Draw the graph of $y = 3e^x$ on the same axes, indicating clearly the axial intercepts, if any. [2]
- (ii) Solve the equation $7 - 3e^{-x} = 3e^x$. [4]

10 The equation of the upper half of a circle is $y = 4 + \sqrt{16 + 6x - x^2}$.

- (i) Find $\frac{dy}{dx}$, giving your answer in the form $\frac{a+bx}{\sqrt{16+6x-x^2}}$, where a and b are constants to be found. [2]

- (ii) Show that the radius of the circle is 5 units and find the coordinates of C , the centre of the circle. [3]

The points P and Q on the circle are such that the tangents to the circle at these two points both have a gradient of $\frac{3}{4}$.

(iii) Find the equations of these two tangents.

[6]

- 11** The temperature, $T^{\circ}\text{C}$, of the coffee in a cup t minutes after boiling water is poured into the cup is given by $T = 28 + Ae^{-kt}$, where A and k are constants.

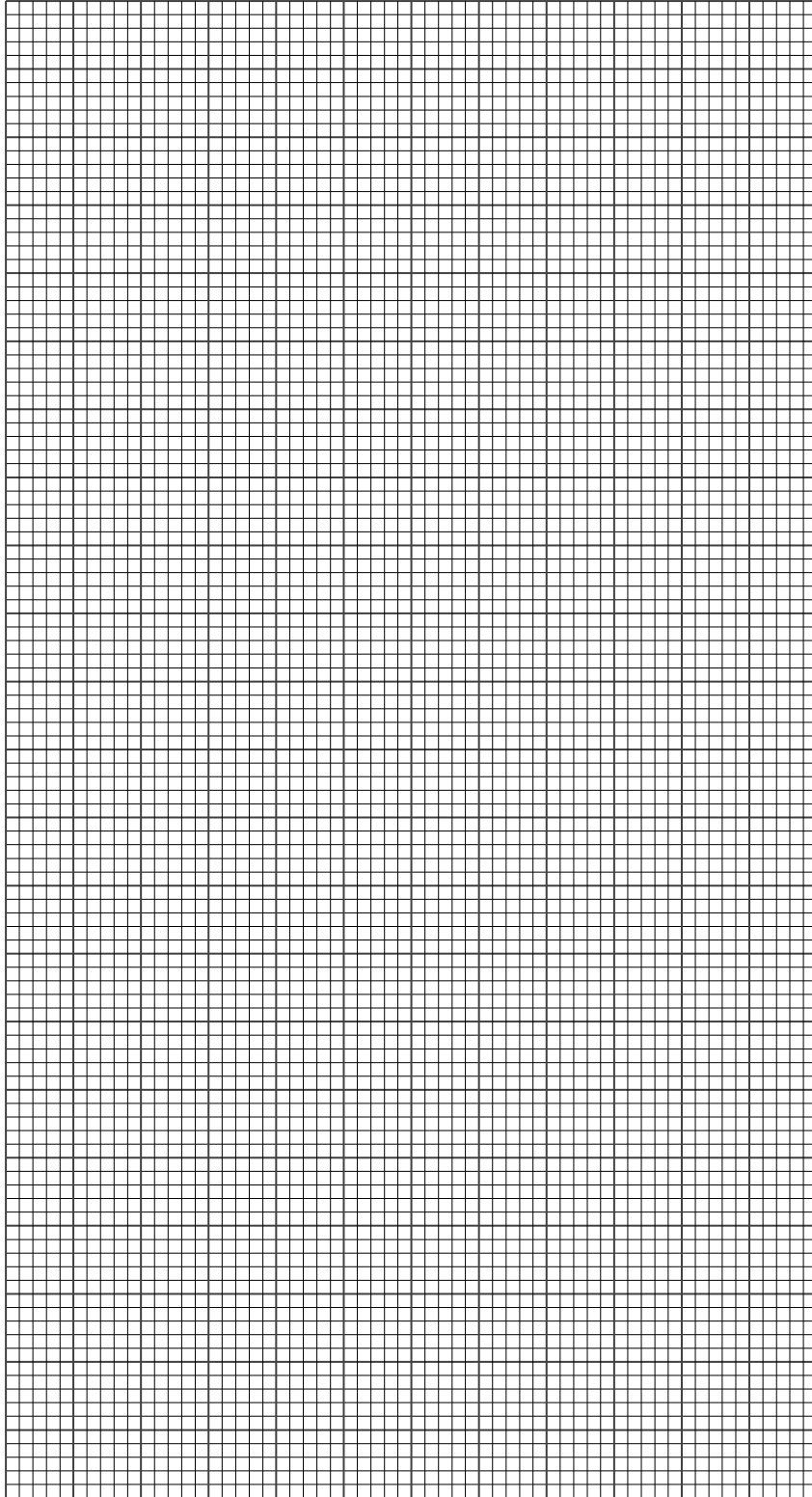
The table below shows corresponding values of t and T .

t	5	10	15	20	25
T	87.0	76.2	67.6	60.3	54.5

- (i)** Plot $\ln(T - 28)$ against t and draw a straight line graph to illustrate the information. [2]

- (ii)** Use your graph to estimate the values of the constants A and k . [4]

- (iii)** Use your graph to estimate how long it takes for the temperature of the coffee to drop to 70°C . [2]



- 12** A particle travels in a straight line and it is initially at the point O . The displacement, s metres from O , of the particle at time t seconds is given $s = te^{-0.1t}$.

(i) Obtain an expression for the velocity of the particle at time t seconds. [2]

(ii) Determine when the particle comes to instantaneous rest. [2]

(iii) Find the distance travelled by the particle in the first 12 seconds. [2]

(iv) Show that the particle does not accelerate in the first 20 seconds.

[4]

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